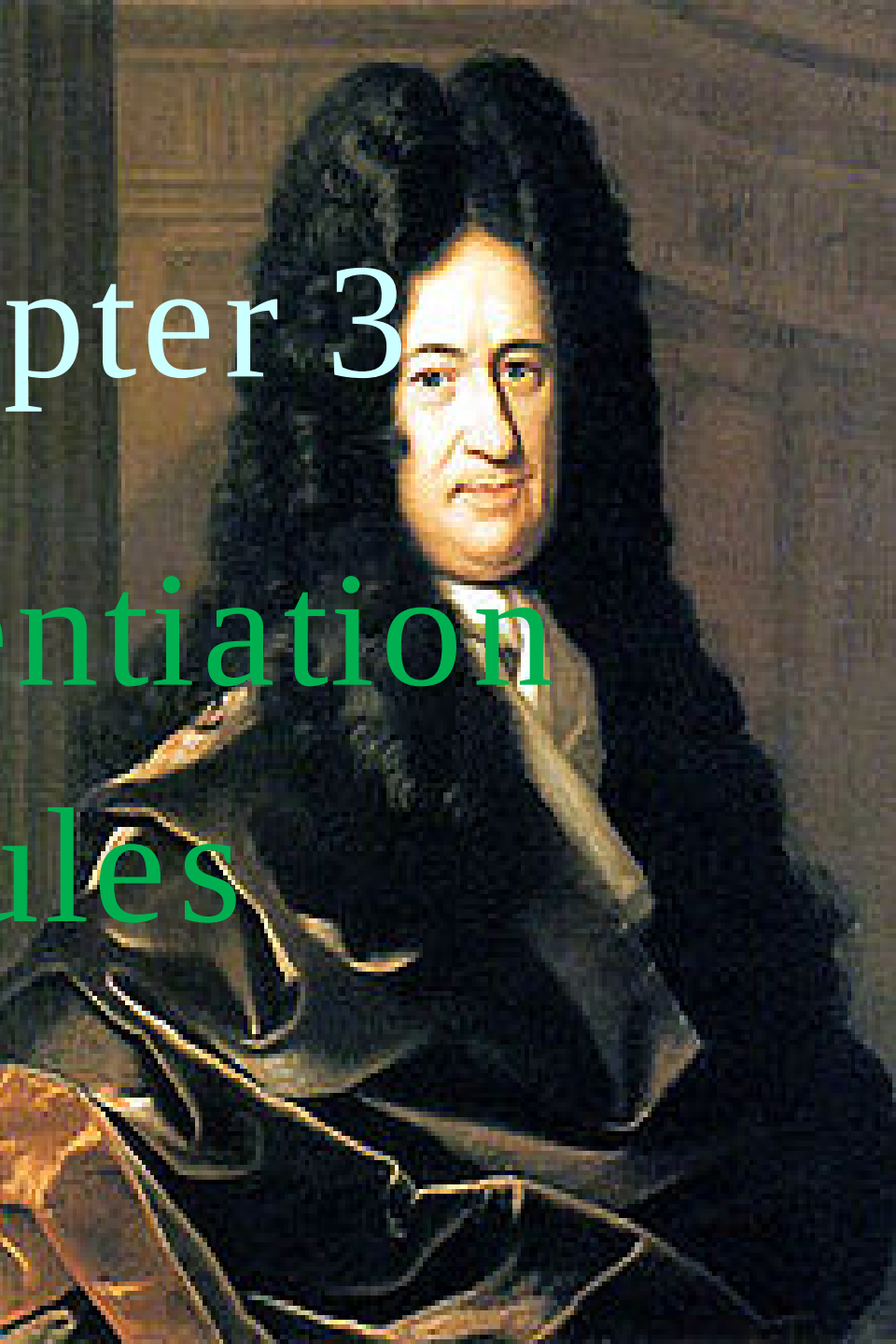




拉曼大學

UCCM1153

Introduction to Calculus and Applications



Chapter 3

Differentiation Rules

The History of Differentiation

Differentiation is part of the science of **Calculus**, and was first developed in the 17th century by two different Mathematicians.



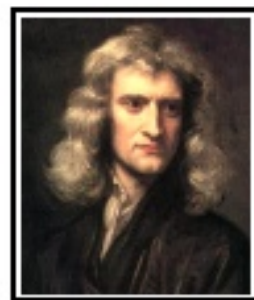
Gottfried Leibniz
(1646-1716)

Germany



Differentiation, or finding the **instantaneous rate of change**, is an essential part of:

- Mathematics and Physics
- Chemistry
- Biology
- Computer Science
- Engineering
- Navigation and Astronomy



Sir Isaac Newton
(1642-1727)

England



Section 3.1

Differentiation Formulas

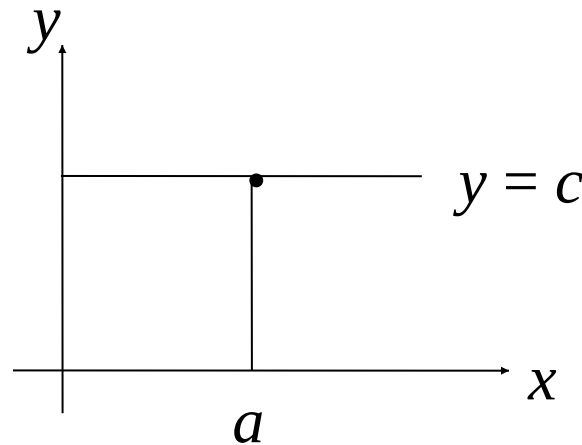
Differentiation Formulas

(1) Derivative of a Constant Function $c, \frac{d}{dx}(c) = 0.$

Proof:

Let $f(x) = c$, from the limit definition,

$$\begin{aligned}\frac{d}{dx}(c) &= f'(x) \\ &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{c - c}{h} \\ &= \lim_{h \rightarrow 0} 0 \\ &= 0\end{aligned}$$



♥ For any constant c , the horizontal line $y = c$ has a tangent line with zero slope.

Differentiation Formulas

(2) **The Power Rule**, $\frac{d}{dx}(x^n) = nx^{n-1}$ for $n \neq 0$.

Example 3.1

Determine the $f'(x)$.

(a) $f(x) = x$; $f'(x) = 1$

(b) $f(x) = x^3$; $f'(x) = 3x^2$

(c) $f(x) = x^{3/2}$; $f'(x) = 3/2x^{1/2}$

(d) $f(x) = x^{-1}$; $f'(x) = -x^{-2}$

Differentiation Formulas

(3) The Constant Multiple Rule $\frac{d}{dx}(cf(x)) = c \frac{d}{dx}f(x) = c'f(x)$

Example 3.2

Determine the $f'(x)$ when $f(x) = 3x^4$ and $f(x) = -x$.

$$\begin{aligned} \text{a } \frac{d}{dx}(3x^4) &= 3 \frac{d}{dx}(x^4) \\ &= 3(x^4)^{1/4} = 3x^3 \end{aligned}$$

$$\begin{aligned} \text{b } \frac{d}{dx}(-x) &= - \left[\frac{d}{dx}(x) \right] \\ &= - \left(1 \right) = -1 \end{aligned}$$

Differentiation Formulas

(4) **The Sum Rule** If f and g are both differentiable, then

$$\frac{d}{dx}(f + g) = \frac{d}{dx}f + \frac{d}{dx}g$$

(5) **The Difference Rule** If f and g are both differentiable, then

$$\frac{d}{dx}(f - g) = \frac{d}{dx}f - \frac{d}{dx}g$$

Differentiation Formulas

Example 3.3

$$\frac{d}{dx} (x^8 + 1x^5 - 2x^4 + 1x^0 - 6x + 5)$$

$$\frac{d}{dx} (x^8 + x^5 - 2x^4 + 1x^0 - 6x + 5)$$

$$= \frac{d}{dx} (x^8) + \frac{d}{dx} (x^5) - \frac{d}{dx} (2x^4) + \frac{d}{dx} (1x^0) - \frac{d}{dx} (6x) + \frac{d}{dx} (5)$$

$$= 8x^7 + 5x^4 - 8x^3 + 0 - 6 + 0$$

$$= 8x^7 + 5x^4 - 8x^3 - 6$$

$$= 8x^7 + 5x^4 - 8x^3 - 6$$

Differentiation Formulas

Example 3.4

Find the points on the curve $y = x^4 - 6x^2 + 4$ where the tangent line is horizontal.

$$\begin{aligned}\frac{d}{dx} (x^4 - 6x^2 + 4) &= 4x^3 - 12x + 0 \\ &= 4x^2(x - 3)\end{aligned}$$

Thus, $dy/dx = 0$ if $x = 0$ or $x^2 - 3 = 0$, that is, $x = \pm \sqrt{3}$.

Differentiation Formulas

(6) **The Product Rule** If f and g are both differentiable, then

$$\frac{d}{dx}(f \cdot g) = f \frac{d}{dx}g + g \frac{d}{dx}f$$

Example 3.5

Find $F'(x)$ if $F(x) = (x^3 - 2x^2 + 3x - 4)^4$.

$$\begin{aligned} F'(x) &= 4(x^3 - 2x^2 + 3x - 4)^3 \cdot \frac{d}{dx}(x^3 - 2x^2 + 3x - 4) \\ &= 4(x^3 - 2x^2 + 3x - 4)^3 (3x^2 - 4x + 3) \\ &= 4 \cdot 6x^6 - 12x^5 + 12x^4 - 48x^3 + 48x^2 - 48x + 48 \\ &= 24x^6 - 12x^5 + 12x^4 - 48x^3 + 48x^2 - 48x + 48 \end{aligned}$$

Differentiation Formulas

Example 3.6

If $h(x) = xg(x)$ and given $g(3) = 5$ and $g'(3) = 2$, find $h'(3)$.

$$\begin{aligned} h(x) &= xg(x) \\ \frac{d}{dx} h(x) &= \frac{d}{dx} [xg(x)] \\ &= x'g(x) + xg'(x) \end{aligned}$$

$$h'(3) = 3'g(3) + 3g'(3) = 1 \cdot 5 + 3 \cdot 2 = 11$$

Differentiation Formulas

(7) **The Quotient Rule** If f and g are both differentiable, then

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x) \frac{d}{dx} [f(x)] - f(x) \frac{d}{dx} [g(x)]}{[g(x)]^2}$$

Differentiation Formulas

Example 3.7

Let $y = \frac{x^2 + x - 2}{x^3 + 6}$, evaluate y' .

$$y' = \frac{\left(\frac{d}{dx}(x^2 + x - 2) \right) - \left(\frac{d}{dx}(x^3 + 6) \right) \left(\frac{x^2 + x - 2}{x^3 + 6} \right)}{(x^3 + 6)^2}$$

$$= \frac{(2x + 1) - (3x^2) \left(\frac{x^2 + x - 2}{x^3 + 6} \right)}{(x^3 + 6)^2}$$

$$= \frac{(2x + 1) - (3x^2 + 3x - 6)}{(x^3 + 6)^2}$$

$$= \frac{-x^2 - 2x + 7}{(x^3 + 6)^2}$$

Differentiation Formulas

$$(8) \quad \frac{d}{dx} e^x = e^x$$

Example 3.8

Find y' if $y = e^x - x$

$$y' = e^x - 1$$

Differentiation Formulas

(9) Equation of the **tangent line at (x_0, y_0)** on the curve $y = f(x)$ is given by ~~$y - y_0 = f'(x_0)(x - x_0)$~~ .

Example 3.9

Find the equation of the tangent line at $(0, 2)$ on the curve $y = x^4 + 2e^x$.

$$y' = 4x^3 + 2e^x$$

$$y'(0) = 2e^0 = 2$$

$$y - 2 = 2(x - 0)$$

$$y = 2x + 2$$

Differentiation Formulas

(10) Equation of the **normal line** at (x_0, y_0) on the curve $y = f(x)$ is given by

$$y - y_0 = -\frac{1}{f'(x_0)}(x - x_0)$$

Example 3.10

Find the equation of the normal line at $(0, 2)$ on the curve $y = x^4 + 2e^x$.

$$y' = 4x^3 + 2e^x$$

$$y'(0) = 2e^0 = 2$$

$$y - 2 = -\frac{1}{2}(x - 0)$$

$$y = -\frac{1}{2}x + 2$$

Differentiation Formulas

$$(11) \frac{d}{dx} \sin x = \cos x$$

Example 3.11

Find $\frac{dy}{dx}$ for $y = x^2 \sin x$

$$\begin{aligned} \frac{d}{dx} x^2 \sin x &= x^2 \frac{d}{dx} (\sin x) + \sin x \frac{d}{dx} x^2 \\ &= x^2 \cos x + 2x \sin x \end{aligned}$$

Differentiation Formulas

$$(12) \quad \frac{d}{dx} \cos x = -\sin x$$

Example 3.12

Find $\frac{dy}{dx}$ for $y = 2 \sin x \cos x$

$$\begin{aligned} y' &= 2[\sin(-\sin) + \cos(\cos)] \\ &= 2[-\sin x + \cos^2 x] \\ &= 2\cos^2 x - 2\sin^2 x \end{aligned}$$

$$\begin{aligned} \cos 2x &= \cos^2 x - \sin^2 x \\ &= 2\cos^2 x - 1 \\ &= 1 - 2\sin^2 x \end{aligned}$$

Differentiation Formulas

$$(13) \quad \frac{d}{dx} \tan x = \frac{d \sin x}{d \cos x} = \sec^2 x$$

$$(14) \quad \frac{d}{dx} \csc x = \frac{d}{d \sin x} \frac{1}{\sin x} = -\csc x \cot x$$

$$(15) \quad \frac{d}{dx} \sec x = \frac{d}{d \cos x} \frac{1}{\cos x} = \sec x \tan x$$

$$(16) \quad \frac{d}{dx} \cot x = \frac{d}{d \tan x} \frac{1}{\tan x} = -\csc^2 x$$

Differentiation Formulas

Proof:

$$\begin{aligned} & \frac{d}{dx}(\tan x) \\ &= \frac{d}{dx} \left(\frac{\sin x}{\cos x} \right) \\ &= \frac{\cos x (\cos x) - \sin x (-\sin x)}{\cos^2 x} \\ &= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} \\ &= \frac{1}{\cos^2 x} = \sec^2 x \end{aligned}$$

$$\begin{aligned} \csc x &= \frac{1}{\sin x} \\ \sec x &= \frac{1}{\cos x} \\ \cot x &= \frac{1}{\tan x} = \frac{\cos x}{\sin x} \\ \tan x &= \frac{\sin x}{\cos x} \end{aligned}$$

Differentiation Formulas

Example 3.13

Differentiate $f(x) = \frac{\sec x}{1 + \tan x}$.

$$\begin{aligned}
 f'(x) &= \frac{\frac{d}{dx}(\sec x) \sec x - (1 + \tan x)}{(1 + \tan x)^2} \\
 &= \frac{(\sec x \tan x) \sec x - (1 + \tan x)}{(1 + \tan x)^2} \\
 &= \frac{\sec^2 x \tan x - (1 + \tan x)}{(1 + \tan x)^2} \\
 &= \frac{\sec^2 x \tan x - 1 - \tan x}{(1 + \tan x)^2}
 \end{aligned}$$

Differentiation Formulas

(17) **The Chain Rule**

If f and g are both differentiable and $f \circ g$ is differentiable, then $(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$.

$$\left[\text{If } y = f(u), u = g(x), \text{ then } \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \right]$$

Note:

- If $y = (g(x))^n$, then $\frac{dy}{dx} = n(g(x))^{n-1} g'(x)$.

Differentiation Formulas

Example 3.14

(a) Find $F'(x)$ if $F(x) = \sqrt{x^2 + 1}$

Let $u = x^2 + 1$ and $y = \sqrt{u}$,

$$F'(x) = \frac{dy}{du} \cdot \frac{du}{dx} = \left(\frac{1}{2\sqrt{u}} \right) \cdot \frac{2x}{1} = \frac{x}{\sqrt{x^2 + 1}}$$

Differentiation Formulas

Example 3.14

(b) Differentiate $y = \sin x^2$.

$$\frac{dy}{dx} = \underbrace{\sin(x^2)}_{\text{outer function}} \underbrace{\cos(x^2)}_{\text{derivative of outer function}} \underbrace{2x}_{\text{derivative of inner function}} = 2x \cos(x^2)$$

Let $u = x^2$

$\therefore y = \sin u$

$$\frac{du}{dx} = 2x$$

$$\frac{dy}{du} = \cos u$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = \cos u (2x) = 2x \cos(x^2).$$

Differentiation Formulas

Example 3.14

(c) Differentiate ~~$y = 2x^4 + 3x^3 + 1$~~

$$\frac{dy}{dx} = \left(\frac{d}{dx} 2x^4 + \frac{d}{dx} 3x^3 + \frac{d}{dx} 1 \right)$$

$$= \left(2 \cdot 4x^{4-1} + 3 \cdot 3x^{3-1} + 0 \right)$$

$$= 8x^3 + 9x^2 + 0$$

$$= 8x^3 + 9x^2$$

$$= 8x^3 + 9x^2$$

$$\frac{dy}{dx} = 8x^3 + 9x^2$$

Differentiation Formulas

Example 3.14

(d) Differentiate $y = \tan x^2 + 1$.

$$\text{Let } u = x^2 + 1$$

$$\frac{d}{dx} [\tan (x^2 + 1)]$$

$$= \frac{d}{dx} [\tan u]$$

$$= \sec^2 u \frac{du}{dx}$$

$$= \sec^2 (x^2 + 1) \cdot \frac{d}{dx} [x^2 + 1]$$

$$= \sec^2 (x^2 + 1) \cdot 2x$$

$$= 2x \sec^2 (x^2 + 1)$$

Differentiation Formulas

$$(18) \quad \frac{d}{dx} a^x = a^x \ln a, \quad a > 0 \quad \left[\frac{d}{dx} (e^x) = \frac{d}{dx} (e^x \ln e) = e^x \right]$$

Note:

- By applying the chain rule, $\frac{d}{dx} e^{g(x)} = e^{g(x)} g'(x)$.

Example 3.15

Differentiate $e^{\sin x}$.

$$\frac{d}{dx} e^{\sin x} = e^{\sin x} \cos x$$

Differentiation Formulas

Extension to the Chain Rule

Suppose $y = f(u)$, $u = g(x)$ and $x = h(t)$, and f , g and h are differentiable, then $\frac{dy}{dt} = \frac{dy}{du} \cdot \frac{du}{dx} \cdot \frac{dx}{dt}$

Example 3.16

Let ~~$f(x) = \sin(\cos(\tan x))$~~ , find $f'(x)$.

$$\begin{aligned} f'(x) &= \cos(\cos(\tan x)) \frac{d}{dx} (\cos(\tan x)) \\ &= \cos(\cos(\tan x)) (-\sin(\tan x)) \frac{d}{dx} (\tan x) \\ &= -\cos(\cos(\tan x)) \sin(\tan x) \sec^2 x \end{aligned}$$

Section 3.2

Implicit Differentiation

Implicit Differentiation

➤ To find $\frac{dy}{dx}$ without solving for y .

Steps:

- First differentiate both sides of the equation with respect to x ;
- Then solve the resulting equation for y' .

Implicit Differentiation

Example 3.17

$$(a) \ y = 0, \frac{dy}{dx} = 0.$$

$$(b) \ y^2 = 0,$$

$$2 \frac{dy}{dx} y = 0$$

$$\frac{dy}{dx} = 0$$

$$(c) \ xy = 0$$

$$y + x \frac{dy}{dx} = 0$$

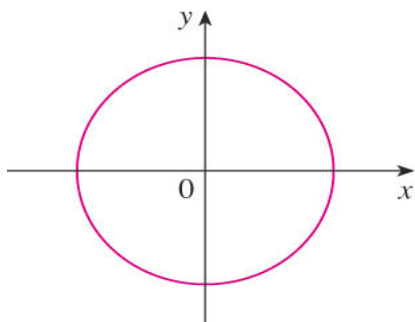
$$\frac{dy}{dx} = - \frac{y}{x}$$

Implicit Differentiation

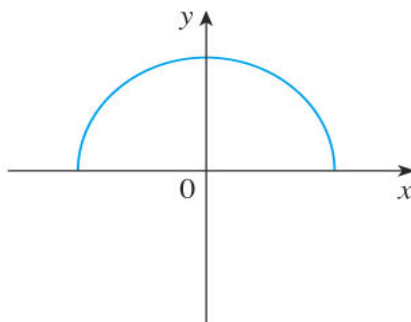
Example 3.18

Find $\frac{dy}{dx}$ and equation of the tangent line at $(3, 4)$ for the circle

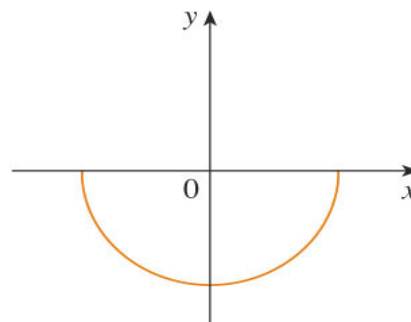
$$x^2 + y^2 = 25$$



(a) $x^2 + y^2 = 25$



(b) $f(x) = \sqrt{25 - x^2}$



(c) $g(x) = -\sqrt{25 - x^2}$

Implicit Differentiation

Example 1.38(Cont')

$$\sqrt{25x^2}$$

$$\frac{d}{dx}(x^2y^2) = (2x \cdot 2y^2)$$

$$\frac{d}{dx}(x^2y^2) = (2x \cdot 2y^2) = 0$$

$$\frac{d}{dx}(y^2) = \frac{d}{dy}(y^2) \cdot \frac{dy}{dx} = 2y \cdot \frac{dy}{dx}$$

$$2y \cdot \frac{dy}{dx} = \frac{d}{dx} y$$

$$\frac{dy}{dx} = -\frac{2x}{2y}$$

$$f'(x,y) = \frac{1}{2} (2x^2y^2)^{-1/2} \frac{d}{dx} (2x^2y^2)$$

$$= \frac{1}{2} (2x^2y^2)^{-1/2} (2x \cdot 2y^2)$$

$$= \frac{x}{\sqrt{25x^2}}$$

$$f'(3) = \frac{3}{\sqrt{25 \cdot 3^2}} = \frac{3}{4}$$

An equation of the tangent is $3x + 4y = 25$.

Implicit Differentiation

Example 3.19

Find $\frac{dy}{dx}$ and the points on the curve $x^3 + y^3 = 6xy$ at which the tangent line is horizontal.

Differentiating both sides of $x^3 + y^3 = 6xy$ with respect to x , regarding y as a function of x , and using the Chain Rule on y^3 and the Product Rule on $6xy$, we get:

$$\begin{aligned} 3x^2 + 3y^2y' &= 6xy' + 6y \\ \text{or} \quad x^2 + y^2y' &= 2xy' + 2y \end{aligned}$$

Implicit Differentiation

Example 1.39(Cont')

Solved for y' : $y^2(2-x) = y^2 - x$

$$(y^2 - x) y' = y^2 - x$$

$$y' = \frac{2yx - x^2}{y^2 - x}$$

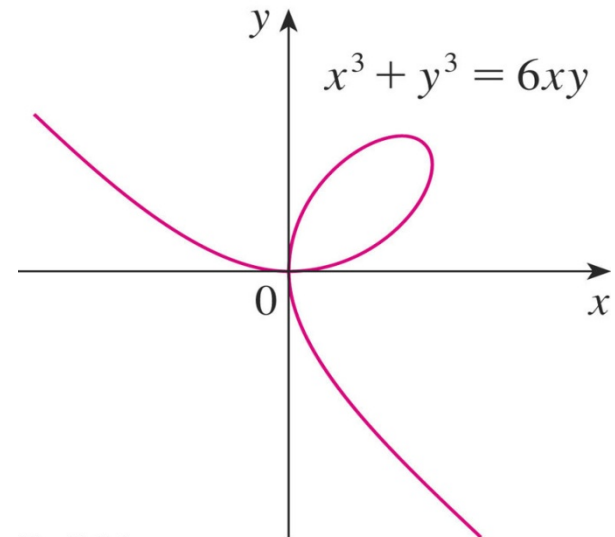
The tangent line is horizontal if $y' = 0$.

Using the expression from the previous result y' , we see that $y' = 0$ when $2y - x^2 = 0$ (provided that $y^2 - x \neq 0$).

Substituting $y = \frac{1}{2}x^2$ in the equation of the curve, we get $x^3 + (\frac{1}{2}x^2)^3 = 6x(\frac{1}{2}x^2)$ which simplifies to $x^6 = 16x^3$.

If $x = 16^{1/3} = 2^{4/3}$, then $y = \frac{1}{2}(2^{8/3}) = 2^{5/3}$.

Thus, the tangent is horizontal at $(2^{4/3}, 2^{5/3})$, which is approximately (2.5198, 3.1748).



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Implicit Differentiation

Example 3.20

Find $\frac{dy}{dx}$ if $\sin y^2 \cos$

Differentiating implicitly with respect to x and remembering that y is a function of x , we get:

$$\cos(y^2) \cdot (1) \cdot (\sin)(\cos)(2y)$$

$$\cos(y^2) \sin(2y \cos y) \cos(y)$$

$$y' = \frac{y^2 \sin x \cos y}{2y \cos x \cos y}$$

Section 3.3

Inverse Functions

Inverse Functions

(1) Inverse Trigonometric Functions

(a)
$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}, -1 < x < 1$$

Prove:

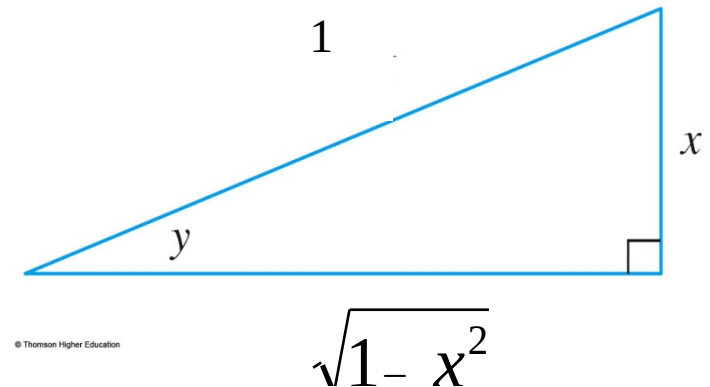
Differentiating $\sin y = x$ implicitly with respect to x , we obtain:

$$\cos y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\cos y}$$

$\cos y \geq 0$ since $-\pi/2 \leq y \leq \pi/2$, so

Therefore
$$\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-x^2}}$$



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Inverse Functions

(1) Inverse Trigonometric Functions

(b) ~~$y = \cos^{-1} x$~~

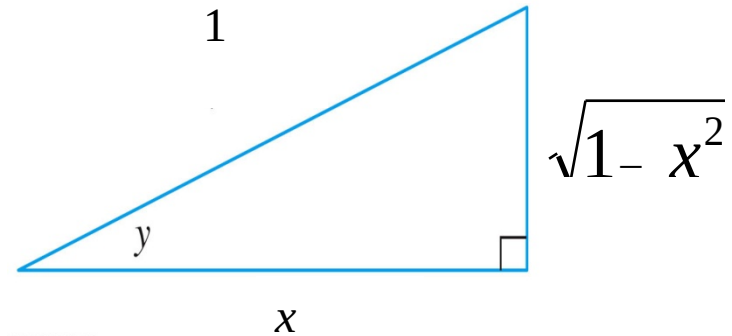
$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}, -1 < x < 1$$

Prove:

~~$$\cos x$$~~

$$\frac{dy}{dx}(-\sin y) = 1$$

~~$$\frac{dy}{dx} = \frac{1}{\sin y} = \frac{1}{\sqrt{1-x^2}}$$~~



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Inverse Functions

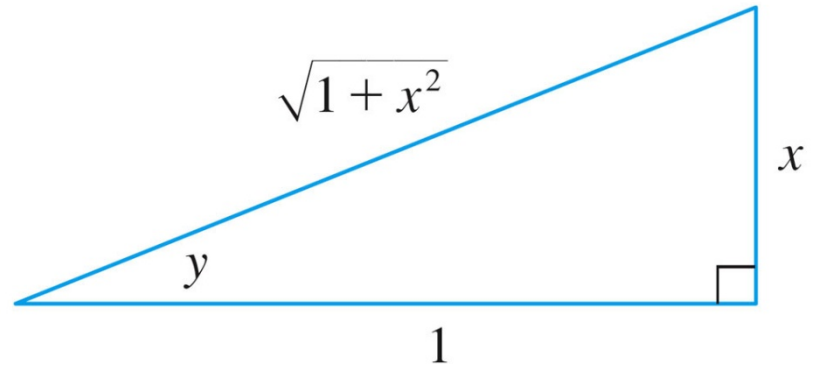
(c) ~~$y = \tan^{-1} x$~~ $y = \tan^{-1} x$ $\frac{\pi}{2} < x < \frac{\pi}{2}$

$$\frac{dy}{dx} = \frac{1}{1+x^2}$$

Prove:

~~$\tan y = x$~~

$$\frac{dy}{dx} (\sec^2 y) = 1$$



~~$\frac{dy}{dx} \frac{1}{\sec^2 y} = \frac{1}{\sec^2 y} = \frac{1}{1+x^2}$~~

Inverse Functions

Example 3.21

Differentiate

a) $y = \frac{1}{\sin^{-1} x}.$

$$\frac{d}{dx} y = \frac{d}{dx} (\sin^{-1} x)$$

$$= - (\sin^{-1} x) \frac{d}{dx} (\sin)$$

$$= - \frac{1}{(\sin^{-1} x)^2 - 1}$$

$$y = \tan^{-1} \sqrt{x}$$

$$f' = \left(\frac{1}{1 + \sqrt{x}} \right) \cdot \frac{1}{2} x^{-1/2} \tan^{-1} \sqrt{x} \\ = \frac{\sqrt{x}}{2(1 + \sqrt{x})} \arctan \sqrt{x}$$

Inverse Functions

(2) Logarithmic Functions

$$y = \log_a x \Rightarrow \frac{dy}{dx} = \frac{1}{x \ln a}, a \neq 0$$

$$\log_a a = \frac{\ln a}{\ln a}$$

$$\frac{d}{dx} (\log_a x) = \frac{1}{x \ln a}$$

Special case:

$$y = \ln x \Rightarrow \frac{dy}{dx} = \frac{1}{x}$$

Inverse Functions

Example 3.22

(a) $y = \ln(x^3 + 1)$

Let $u = x^3 + 1$, then, $y = \ln u$.

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} = \frac{1}{u} \frac{du}{dx} \\ &= \frac{1}{x^3 + 1} (3x^2) \\ &= \frac{3x^2}{x^3 + 1} \end{aligned}$$

(b) $y = \ln \sin x$

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} = \frac{1}{\sin x} (\cos x) \\ &= \cot x \end{aligned}$$

Inverse Functions

Example 3.22(Cont')

(c) $y = (\ln x)^{\frac{1}{2}}$

$$f' = \left(\frac{1}{2} \cdot (\ln x)^{-\frac{1}{2}} \right) \frac{d}{dx} (\ln x) = \frac{1}{2\sqrt{\ln x}} \cdot \frac{1}{x} = \frac{1}{2x\sqrt{\ln x}}$$

(d) ~~$f(x) = \log_{10} (2 \sin x)$~~

$$\begin{aligned} \frac{d}{dx} \log_{10} (2 \sin x) &= \frac{1}{(2 \sin x) \ln 10} \frac{d}{dx} (2 \sin x) \\ &= \frac{\cos x}{(2 \sin x) \ln 10} \end{aligned}$$

Note:

~~$$f(x) = \ln x \Rightarrow f'(x) = \frac{1}{x}$$~~

- Logarithmic differentiation is used when involves complicated functions

Inverse Functions

Example 3.23

Differentiate

$$(a) \quad y = \frac{x^{3/4} \sqrt{x^2 + 1}}{(3x + 2)^5}$$

~~$$\ln y = \ln(x^{3/4}) + \ln(\sqrt{x^2 + 1}) - \ln(3x + 2)$$~~

~~$$\frac{1}{y} \frac{dy}{dx} = \frac{3}{4} \cdot \frac{1}{x} + \frac{1}{2} \cdot \frac{2x}{x^2 + 1} - \frac{3}{3x + 2}$$~~

~~$$\frac{d}{dx} \ln y = \frac{3}{4x} + \frac{x}{x^2 + 1} - \frac{3}{3x + 2}$$~~

Inverse Functions

Example 3.23(Cont')

(b) $y = a^{g(x)}$, where a is constant.

$$\frac{d}{dx}[a^{g(x)}] = a^{g(x)}(\ln a)g'(x)$$

(c) $y = [f(x)]^{g(x)}$

$$\ln y = g(x) \ln[f(x)]$$

$$\frac{1}{y} \frac{dy}{dx} = g(x) \frac{1}{f(x)} f'(x) + \ln[f(x)] g'(x)$$

$$\frac{dy}{dx} = [f(x)]^{g(x)} \left[g(x) \frac{f'(x)}{f(x)} + g'(x) \ln[f(x)] \right]$$

(d)

$$\frac{d}{dx}(x^{\sqrt{x}}) = x^{\sqrt{x}} \left(\frac{1}{x} + \frac{1}{2\sqrt{x}} \right) = x^{\sqrt{x}} \left(\frac{2 + \sqrt{x}}{2\sqrt{x}} \right)$$

Section 3.4

Linear Approximation and Differentials

Linear Approximation and Differentials

(1) Linear Approximation

Recall:

The equation of the tangent line at $(a, f(a))$ on the curve $y = f(x)$ is



When x is near a , the curve $y = f(x)$ can be approximated by the equation of the tangent line at $(a, f(a))$, i.e.



The linear function whose graph is this tangent line, that is  is called the linearization of f at a .

Linear Approximation and Differentials

Example 3.24

Find the linearization of the function $f(x) = \sqrt{x+3}$ at $a = 1$. Use it to approximate $\sqrt{3.98}$ and $\sqrt{4.05}$

$$f(x) = \frac{1}{2\sqrt{x+3}}$$

So, we have $f(1) = 2$ and $f'(1) = 1/4$.

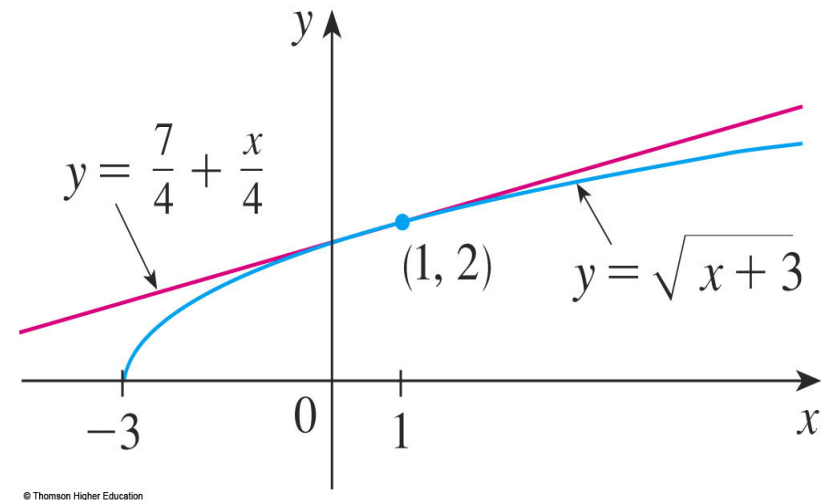
$$L(x) = f(1) + f'(1)(x-1)$$

$$= 2 + \frac{1}{4}(x-1)$$

$$= \frac{7}{4} + \frac{x}{4} \quad (\text{When } x \text{ near to } 1)$$

$$\sqrt{3.98} \approx \frac{7}{4} + \frac{0.98}{4} = 1.995$$

$$\sqrt{4.05} \approx \frac{7}{4} + \frac{1.05}{4} = 2.0125$$



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We see that:

- The tangent line approximation is a good approximation to the given function when x is near 1.
- Our approximations are overestimates, because the tangent line lies above the curve.

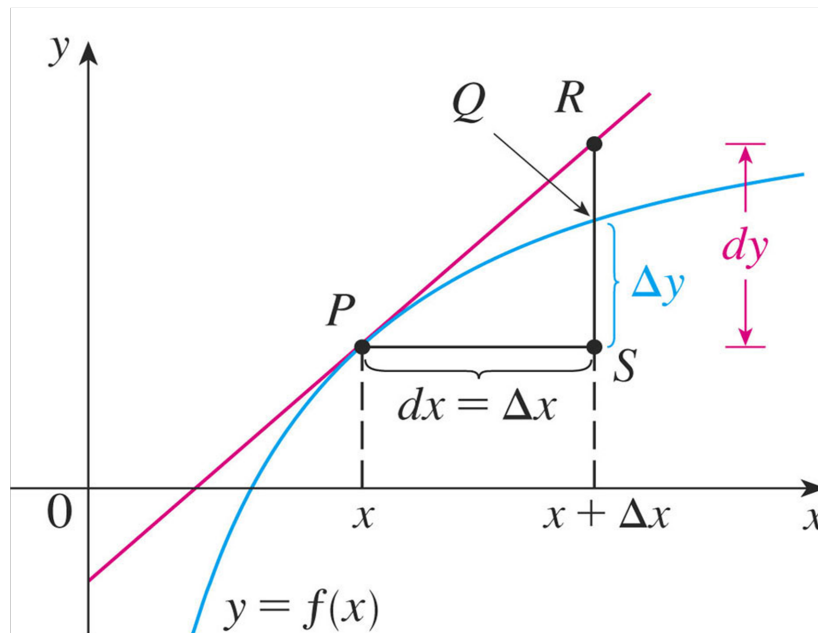
Linear Approximation and Differentials

(2) Differentials

If $y = f(x)$ is differentiable, then the differential

$$dy = f'(x)dx \quad (1)$$

where dx is an independent variable and dy is a dependent variable.



Linear Approximation and Differentials

- Let $P(x, f(x))$ and $Q(x + \Delta x, f(x + \Delta x))$ be points on the graph of f and let $dx = \Delta x$.
- The corresponding change in y is $\Delta y = f(x + \Delta x) - f(x)$
- The slope of the tangent line PR is the derivative $f'(x)$, and so the directed distance from S to R is $f'(x)dx = dy$.
- Thus, dy represents the amount that the tangent line rises or falls (the change in the linearization), whereas Δy is the amount that the curve $y = f(x)$ rises or falls when x changes by an amount dx .
- Note from Fig. 6 that the approximation $\Delta y \approx dy$ improves as Δx becomes smaller.
- If we let $dx = x - a$, then $x = a + dx$ and we can rewrite our linear approximation in the notation of differentials:

$$f(a + dx) \approx f(a) + dy$$

$$\approx f(a) + f'(a)dx$$

Linear Approximation and Differentials

Example 3.25

Given $f(x) = \sqrt{x+3}$, use differential to estimate $\sqrt{4.05}$.

Let $dx = x - a$

$$x = a + dx$$

$$f(x) \approx f(a) + f'(a)(x-a)$$

in the notation of differentials

$$f(a+dx) \approx f(a) + dy$$

$$f(x) = \sqrt{x+3}$$

$$\therefore f'(x) = \frac{1}{2}(x+3)^{-\frac{1}{2}} = \frac{1}{2\sqrt{x+3}}$$

$$\frac{dy}{dx} = f'(x) \Rightarrow dy = f'(x)dx$$

$$dy = \frac{dx}{2\sqrt{x+3}}$$

$$\sqrt{4.05} = \sqrt{1.05+3}, x=1.05$$

where nearest to a , $a=1$

$$dx = x - a = 1.05 - 1 = 0.05$$

$$dy = \frac{0.05}{2\sqrt{1+3}} = 0.0125$$

$$f(a + dx) \approx f(a) + dy$$

$$\sqrt{4.05} \approx f(1.05)$$

$$\approx f(1) + dy$$

$$\approx \sqrt{1+3} + 0.0125$$

$$\approx 2.0125$$

Linear Approximation and Differentials

Example 3.26

The radius of a sphere was measured and found to be 21cm with a possible error in measurement of at most 0.05cm. What is the maximum error in using this value of the radius to compute the volume of the sphere?

If the radius of the sphere is r , then its volume is $V = 4/3\pi r^3$.

If the error in the measured value of r is denoted by $dr = \Delta r$, then the corresponding error in the calculated value of V is ΔV .

This can be approximated by the differential

$$dV = 4\pi r^2 dr$$

When $r = 21$ and $dr = 0.05$, this becomes:

$dV = 4\pi(21)^2 0.05 \approx 277$. The maximum error in the calculated volume is about 277 cm³.